TestBoard_ADC124
Grouped MUX scan: settling and VMID

13 October 2026 | Production firmware | Manual conversion | LPSPI | 16 MHz SPI

Setup and test matrix

Both array connectors disconnected. Laptop charger connected; a common quiet-period mask screens for interference bursts. Two full arrays, 25 sensor routes per array, 50 transmitted values per sweep. Each ADC is an ADC124S101, with four external MUX/op-amp inputs. No sensor strips were connected or pressed.

Parameter	Tested value
Settling delay	1, 2, 5, 10, 50 us
VMID insertion	Off and on at every settling delay
Acquisition	Manual / channel repeat = 1
Scan order	MUX address grouped: array, external input, ADC IN
SPI	LPSPI on one shared bus; requested clock 16 MHz
Duration	3 repetitions x (1 s warm-up + 15 s measurement)
Reference / VMID	Fixed nominal 3.3 V / 1.65 V

Array / ADC	MUX1 / IN1	MUX2 / IN2	MUX3 / IN3	MUX4 / IN4
A / ADC1	470 kOhm	1 MOhm	470 kOhm	1 MOhm
B / ADC2	1 MOhm	470 kOhm	1 MOhm	470 kOhm

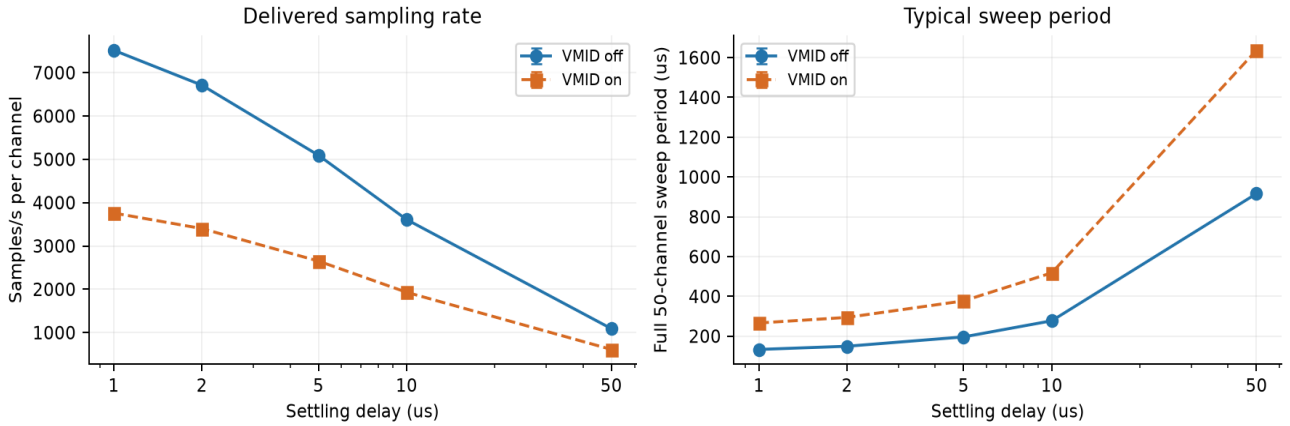
The resistors bias each shared MUX-output/op-amp node toward VMID; they are not separate resistors on every sensor route. Array A uses TMUX1108; B uses TMUX1308A. Array and resistor comparisons therefore include different physical circuits.

Every capture uses scanorder mux. Switch/settle once for each external address, then read ADC IN1-IN4 at addresses 0-5; address 6 reads IN4 alone. This gives 25 values per array, 50 per full sweep. Optional VMID-on parks and converts all four ADC inputs after each group; those conversions are discarded. Mandatory VMID parking remains active in both settings. This report replaces the earlier channel-order characterization for the intended scan plan.

Voltage conversion: V = counts x 3.3/4096. One count = 0.000805664 V. Voltages use the configured nominal reference, without a fresh voltage calibration. The requested SCLK was not measured with an oscilloscope in this test.

Sampling rate and acquisition time

Settle us	VMID	Received Hz / channel	50-channel values/s	Acquisition us	Period us	Overhead us
1	off	7,518.1	375,907	132.0	133.0	1.0
1	on	3,759.4	187,970	265.0	266.0	1.0
2	off	6,711.4	335,570	147.0	149.0	2.0
2	on	3,401.4	170,068	293.0	294.0	1.0
5	off	5,094.1	254,707	195.0	196.0	1.0
5	on	2,651.3	132,563	376.0	377.0	1.0
10	off	3,610.1	180,505	276.0	277.0	1.0
10	on	1,930.5	96,525	517.0	518.0	1.0
50	off	1,091.4	54,568	915.0	916.0	1.0
50	on	610.8	30,540	1636.0	1637.0	1.0



Rates are delivered sweeps per device sweep-start elapsed time after the 1 s warm-up; each sweep contains one value per channel. Total values/s = rate x 50, excluding discarded priming and VMID conversions. Table entries are medians of three repetitions. Error bars show the repetition min-max range.

Acquisition and period use timestamp medians; overhead uses median(period minus acquisition), so independently summarized columns may not add exactly. Received rate counts retained USB frames. Firmware sampling/sent/discarded accounting, including warm-up, is reported separately in the integrity section.

Quiet-period selection and retained data

Warm-up is excluded. A common time mask is applied to all 50 channels, so noise and baseline comparisons use exactly the same quiet intervals within each capture. No individual values are clipped, smoothed or detrended.

For each of eight MUX / ADC-input groups, calculate the median of its channel sample standard deviations in complete 50 ms bins. The bin width gives adequate samples even at the slowest tested sweep rate. Threshold = max(0.60 counts, P20 + 0.15 counts, median of lower 40% + 4 x 1.4826 x MAD of that lower subset). A bin above any group threshold is rejected, with one neighboring bin on each side.

The 0.60-count floor prevents normal toggling between two adjacent ADC codes from being classified as a burst. This is a quantization-aware adaptation of the earlier ADS7953 report method, applied equally to every setting.

This is an operational definition of quiet periods. The laptop charger was connected, but neither charger switching nor rail voltage was recorded. The rejected intervals are interference/activity periods; the measurements do not prove that every burst came from the charger. Persistent individual-channel variation remains in the results.

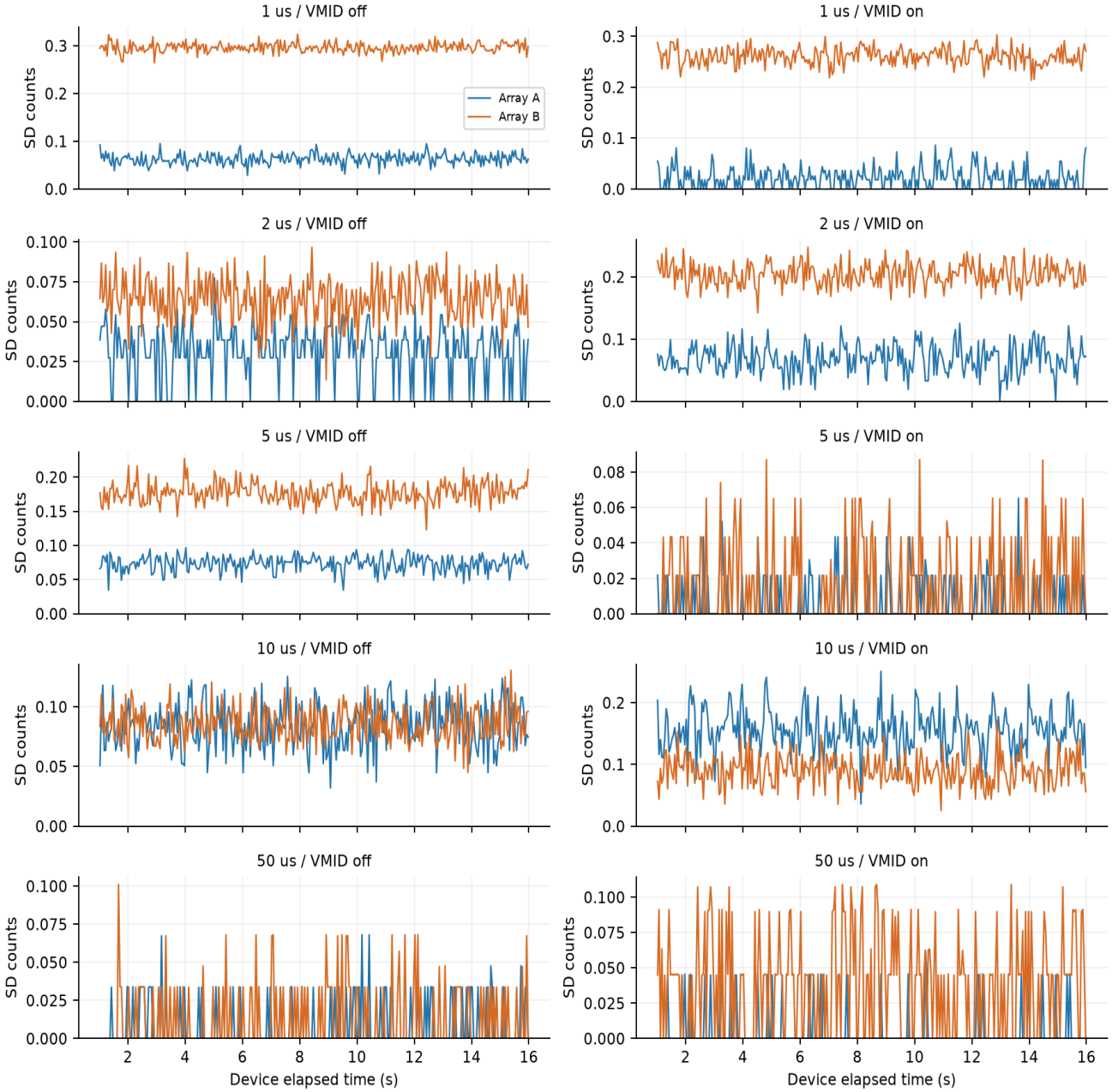
Settle us	VMID	Quiet seconds: median [range]	Rejected time %	Min quiet frames / channel
1	off	15.00 [15.00-15.00]	0.0	112,761
1	on	15.00 [15.00-15.00]	0.0	56,388
2	off	15.00 [15.00-15.00]	0.0	100,668
2	on	15.00 [15.00-15.00]	0.0	51,019
5	off	15.00 [15.00-15.00]	0.0	76,409
5	on	15.00 [15.00-15.00]	0.0	39,766
10	off	15.00 [15.00-15.00]	0.0	54,149
10	on	15.00 [15.00-15.00]	0.0	28,956
50	off	15.00 [15.00-15.00]	0.0	16,369
50	on	15.00 [15.00-15.00]	0.0	9,162

All 30 bin masks, eight thresholds per capture and channel metrics are retained alongside the raw data. Peak-to-peak depends on record length and sample count. The repetition CSV includes actual quiet sample counts; the summary CSV includes repetition ranges.

No post-warm-up bin exceeded the quantization-aware burst thresholds in any capture. All 15 s measurement windows were retained. The pronounced charger-burst pattern seen in the earlier board report was not observed here at this ADC resolution and group-envelope criterion. Sub-LSB interference or isolated channel events are not ruled out.

Quiet and burst envelopes

First repetition of every setting. Curves show the median bin SD across the four MUX groups in each array; selection itself uses all eight group envelopes. Shaded bins were excluded from every channel. Group thresholds are saved in analysis.json and bin envelopes in CSV.



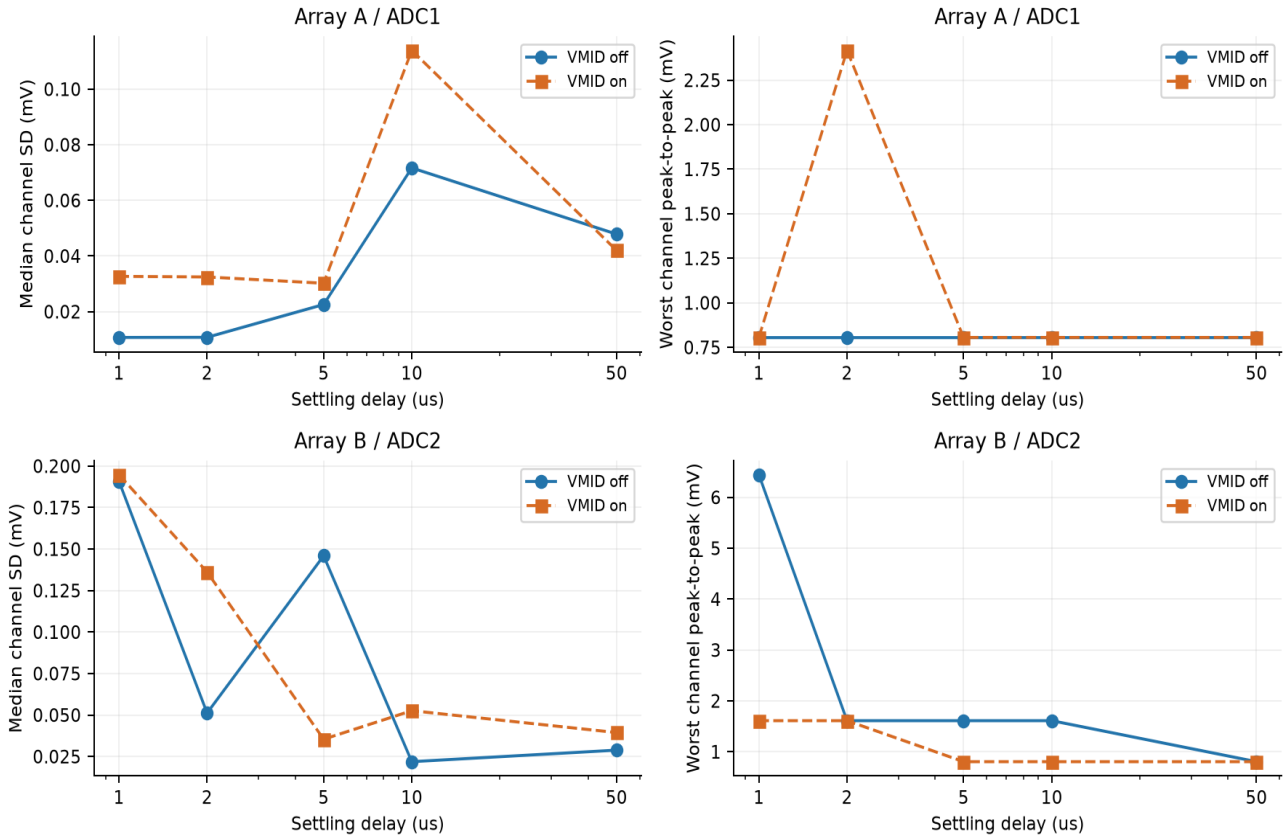
Quiet noise in counts and volts

Each cell shows **median / maximum across channels**, after taking the median of three repetitions per channel. SD uses ddof=1. Peak-to-peak is the actual maximum minus minimum of retained quiet samples, with no individual outlier rejection.

Settle us	VMID	Array	SD counts med / max	SD V med / max	P-P counts med / max	P-P V med / max
1	off	A	0.013 / 0.434	0.000011 / 0.000350	1.0 / 1.0	0.000806 / 0.000806
1	off	B	0.237 / 0.891	0.000191 / 0.000718	2.0 / 8.0	0.001611 / 0.006445
1	on	A	0.041 / 0.500	0.000033 / 0.000403	1.0 / 1.0	0.000806 / 0.000806
1	on	B	0.242 / 0.493	0.000195 / 0.000397	1.0 / 2.0	0.000806 / 0.001611
2	off	A	0.013 / 0.476	0.000011 / 0.000384	1.0 / 1.0	0.000806 / 0.000806
2	off	B	0.064 / 0.500	0.000051 / 0.000403	1.0 / 2.0	0.000806 / 0.001611
2	on	A	0.040 / 0.553	0.000032 / 0.000445	1.0 / 3.0	0.000806 / 0.002417
2	on	B	0.169 / 0.499	0.000136 / 0.000402	1.0 / 2.0	0.000806 / 0.001611
5	off	A	0.028 / 0.497	0.000023 / 0.000400	1.0 / 1.0	0.000806 / 0.000806
5	off	B	0.181 / 0.497	0.000146 / 0.000401	1.0 / 2.0	0.000806 / 0.001611
5	on	A	0.038 / 0.498	0.000030 / 0.000401	1.0 / 1.0	0.000806 / 0.000806
5	on	B	0.044 / 0.448	0.000035 / 0.000361	1.0 / 1.0	0.000806 / 0.000806
10	off	A	0.089 / 0.488	0.000072 / 0.000393	1.0 / 1.0	0.000806 / 0.000806
10	off	B	0.027 / 0.494	0.000022 / 0.000398	1.0 / 2.0	0.000806 / 0.001611
10	on	A	0.141 / 0.500	0.000114 / 0.000403	1.0 / 1.0	0.000806 / 0.000806
10	on	B	0.065 / 0.488	0.000053 / 0.000393	1.0 / 1.0	0.000806 / 0.000806
50	off	A	0.059 / 0.500	0.000048 / 0.000403	1.0 / 1.0	0.000806 / 0.000806
50	off	B	0.036 / 0.467	0.000029 / 0.000376	1.0 / 1.0	0.000806 / 0.000806
50	on	A	0.052 / 0.365	0.000042 / 0.000294	1.0 / 1.0	0.000806 / 0.000806
50	on	B	0.049 / 0.464	0.000039 / 0.000374	1.0 / 1.0	0.000806 / 0.000806

Noise is time variation around each channel's own mean. Slow variation between retained quiet periods contributes to SD and peak-to-peak. Baseline spread, on subsequent pages, is variation between channel median levels. Zero SD indicates a constant quantized output code during the record; it does not establish zero analog noise.

Quiet noise versus settling delay



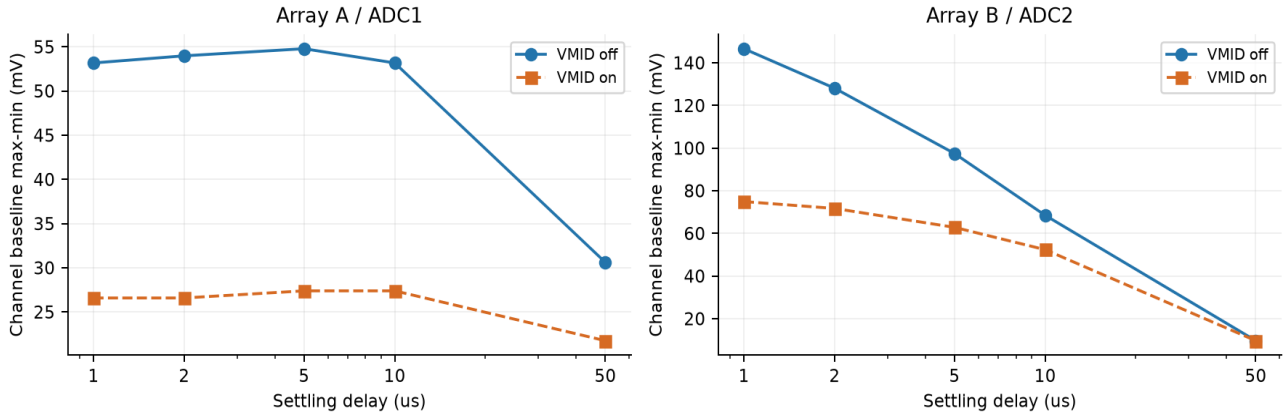
SD graphs show the median channel noise; peak-to-peak graphs show the worst channel. Solid circles: VMID off. Dashed squares: VMID on. Graphs use mV for readability; the table and CSV supply counts and volts.

Longer settling changes both conversion history and achievable rate. Open connectors and the existing bias network make these useful baseline measurements, but they do not demonstrate performance with sensor strips attached or pressed.

Baseline spread between sampled channels

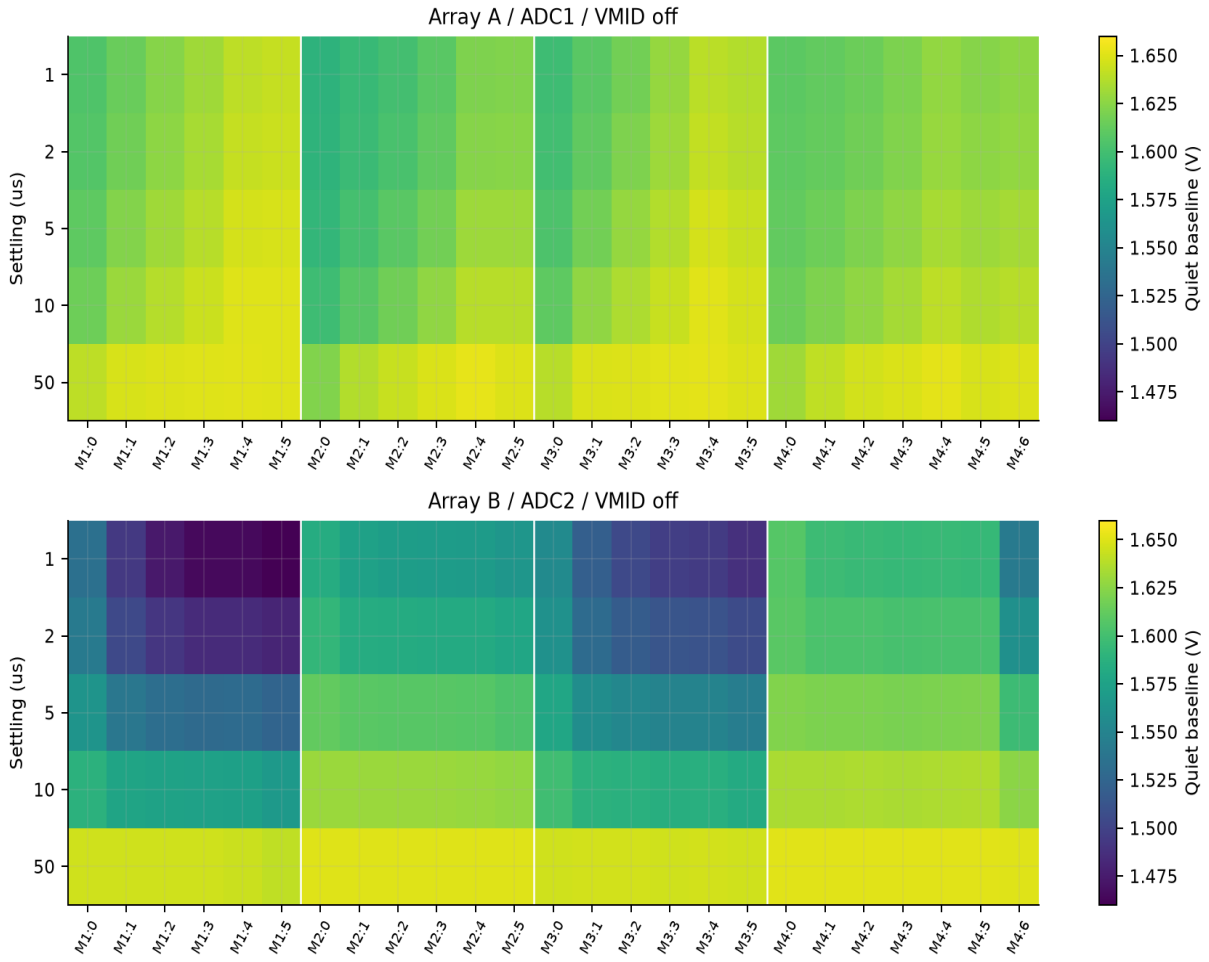
Baseline = median of quiet samples per channel, then median across repetitions. Max-min compares the highest and lowest of 25 channel baselines within one array / ADC. Baseline SD is population SD across those 25 baselines (ddof=0), not time-domain noise.

Settle us	VMID	Array	Max-min counts	Max-min V	Baseline SD counts	Baseline SD V
1	off	A	66.0	0.053174	17.59	0.014174
1	off	B	182.0	0.146631	60.22	0.048521
1	on	A	33.0	0.026587	9.65	0.007778
1	on	B	93.0	0.074927	32.49	0.026174
2	off	A	67.0	0.053979	17.85	0.014379
2	off	B	159.0	0.128101	55.23	0.044501
2	on	A	33.0	0.026587	9.70	0.007818
2	on	B	89.0	0.071704	31.81	0.025625
5	off	A	68.0	0.054785	18.12	0.014599
5	off	B	121.0	0.097485	44.13	0.035554
5	on	A	34.0	0.027393	9.56	0.007705
5	on	B	78.0	0.062842	29.25	0.023565
10	off	A	66.0	0.053174	17.45	0.014060
10	off	B	85.0	0.068481	31.77	0.025597
10	on	A	34.0	0.027393	9.77	0.007869
10	on	B	65.0	0.052368	24.53	0.019764
50	off	A	38.0	0.030615	8.76	0.007055
50	off	B	12.0	0.009668	3.54	0.002849
50	on	A	27.0	0.021753	9.55	0.007696
50	on	B	12.0	0.009668	3.66	0.002948



Every channel baseline - VMID off

Each column is one physical route; Mx:y means MUX x / ADC INx, external input y. Each row is a settling delay. Both panels use the same voltage scale. Values are quiet channel medians, summarized across three repetitions.

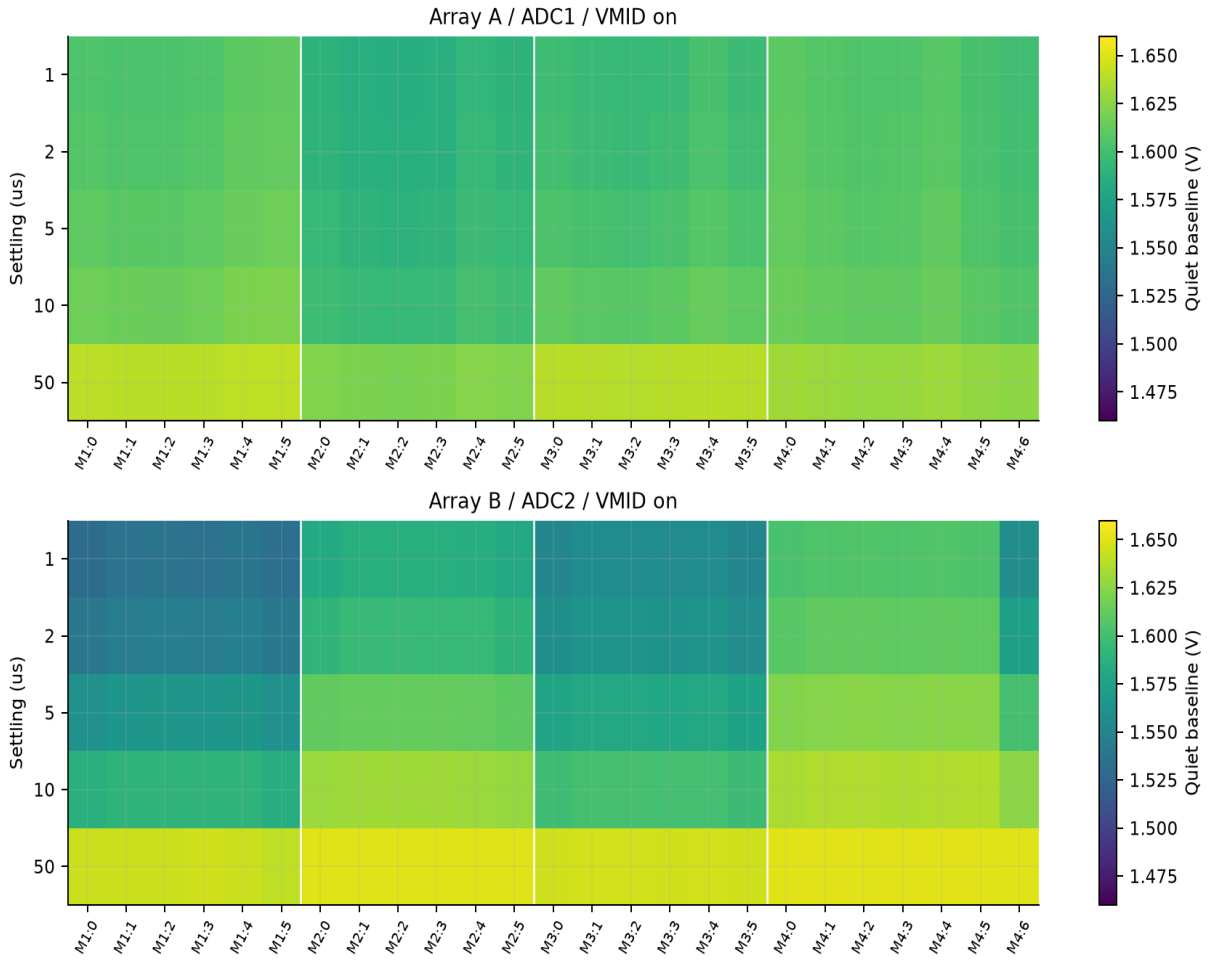


MUX1-3 have six populated sensor routes (inputs 0-5); MUX4 has seven (0-6). Inputs 0-4 map to PZT1, PZT3, PZT5 and PZT6 on MUX1-4 respectively. Input 5 maps to PZT7 channels 1-4; MUX4 input 6 maps to PZT7 channel 5. VMID parking input 7 is not transmitted.

Vertical white lines separate the four ADC-input / MUX groups. Full baseline values in volts/counts, per-channel noise and repetition ranges are in per_channel_metrics.csv.

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Per ADC-input / MUX summary - VMID off

Each row groups the 6 or 7 sensor routes entering one ADC input. Bias is the MUX/op-amp node resistor. SD is median channel noise; P-P is maximum channel peak-to-peak. Spread compares quiet channel baselines within this input group. Group_summary.csv contains all values in both volts and counts.

Settle us	Array	MUX / IN	Bias ohm	Noise SD counts	Worst P-P counts	Spread counts	Spread V	Center V
1	A	1	470k	0.093	1.0	47.0	0.037866	1.627441
1	A	2	1M	0.061	1.0	40.0	0.032227	1.604480
1	A	3	470k	0.177	1.0	52.0	0.041895	1.623010
1	A	4	1M	0.009	1.0	22.0	0.017725	1.620190
1	B	1	1M	0.353	5.0	91.0	0.073315	1.469128
1	B	2	470k	0.231	4.0	23.0	0.018530	1.570642
1	B	3	1M	0.042	5.0	84.0	0.067676	1.500549
1	B	4	470k	0.367	8.0	81.0	0.065259	1.594409
2	A	1	470k	0.130	1.0	47.0	0.037866	1.629858
2	A	2	1M	0.004	1.0	42.0	0.033838	1.606897
2	A	3	470k	0.007	1.0	53.0	0.042700	1.625830
2	A	4	1M	0.056	1.0	23.0	0.018530	1.621802
2	B	1	1M	0.049	2.0	76.0	0.061230	1.488062
2	B	2	470k	0.183	2.0	18.0	0.014502	1.582727
2	B	3	1M	0.072	1.0	69.0	0.055591	1.515454
2	B	4	470k	0.058	1.0	60.0	0.048340	1.602466
5	A	1	470k	0.000	1.0	46.0	0.037061	1.635095
5	A	2	1M	0.151	1.0	47.0	0.037866	1.613342
5	A	3	470k	0.221	1.0	53.0	0.042700	1.632678
5	A	4	1M	0.000	1.0	27.0	0.021753	1.626636
5	B	1	1M	0.261	2.0	49.0	0.039478	1.531165
5	B	2	470k	0.408	2.0	10.0	0.008057	1.608105
5	B	3	1M	0.064	2.0	42.0	0.033838	1.550903
5	B	4	470k	0.076	1.0	31.0	0.024976	1.620190
10	A	1	470k	0.090	1.0	42.0	0.033838	1.641138
10	A	2	1M	0.098	1.0	51.0	0.041089	1.621802
10	A	3	470k	0.062	1.0	50.0	0.040283	1.639526
10	A	4	1M	0.186	1.0	31.0	0.024976	1.633081
10	B	1	1M	0.163	2.0	26.0	0.020947	1.574670
10	B	2	470k	0.189	2.0	3.0	0.002417	1.629858
10	B	3	1M	0.035	1.0	20.0	0.016113	1.586755
10	B	4	470k	0.004	1.0	14.0	0.011279	1.634692
50	A	1	470k	0.004	1.0	13.0	0.010474	1.649597
50	A	2	1M	0.097	1.0	38.0	0.030615	1.645972
50	A	3	470k	0.070	1.0	16.0	0.012891	1.649194
50	A	4	1M	0.022	1.0	25.0	0.020142	1.647583
50	B	1	1M	0.279	1.0	5.0	0.004028	1.645166
50	B	2	470k	0.052	1.0	0.0	0.000000	1.650000
50	B	3	1M	0.049	1.0	1.0	0.000806	1.645569
50	B	4	470k	0.000	1.0	1.0	0.000806	1.650806

Per ADC-input / MUX summary - VMID on

Each row groups the 6 or 7 sensor routes entering one ADC input. Bias is the MUX/op-amp node resistor. SD is median channel noise; P-P is maximum channel peak-to-peak. Spread compares quiet channel baselines within this input group. Group_summary.csv contains all values in both volts and counts.

Settle us	Array	MUX / IN	Bias ohm	Noise SD counts	Worst P-P counts	Spread counts	Spread V	Center V
1	A	1	470k	0.002	1.0	10.0	0.008057	1.604883
1	A	2	1M	0.216	1.0	10.0	0.008057	1.587964
1	A	3	470k	0.053	1.0	10.0	0.008057	1.596021
1	A	4	1M	0.026	1.0	14.0	0.011279	1.604883
1	B	1	1M	0.321	2.0	9.0	0.007251	1.535999
1	B	2	470k	0.460	2.0	6.0	0.004834	1.585144
1	B	3	1M	0.070	2.0	7.0	0.005640	1.556946
1	B	4	470k	0.157	2.0	58.0	0.046729	1.604883
2	A	1	470k	0.051	1.0	9.0	0.007251	1.606494
2	A	2	1M	0.151	1.0	10.0	0.008057	1.588770
2	A	3	470k	0.091	3.0	10.0	0.008057	1.598035
2	A	4	1M	0.040	1.0	14.0	0.011279	1.606494
2	B	1	1M	0.169	1.0	8.0	0.006445	1.544458
2	B	2	470k	0.306	2.0	6.0	0.004834	1.593604
2	B	3	1M	0.297	2.0	7.0	0.005640	1.563391
2	B	4	470k	0.024	1.0	48.0	0.038672	1.610522
5	A	1	470k	0.030	1.0	9.0	0.007251	1.610522
5	A	2	1M	0.009	1.0	10.0	0.008057	1.591992
5	A	3	470k	0.028	1.0	9.0	0.007251	1.602869
5	A	4	1M	0.141	1.0	13.0	0.010474	1.608105
5	B	1	1M	0.026	1.0	6.0	0.004834	1.565405
5	B	2	470k	0.177	1.0	4.0	0.003223	1.612537
5	B	3	1M	0.026	1.0	6.0	0.004834	1.580310
5	B	4	470k	0.044	1.0	29.0	0.023364	1.623413
10	A	1	470k	0.196	1.0	8.0	0.006445	1.616162
10	A	2	1M	0.189	1.0	9.0	0.007251	1.596423
10	A	3	470k	0.026	1.0	7.0	0.005640	1.610120
10	A	4	1M	0.184	1.0	12.0	0.009668	1.611328
10	B	1	1M	0.048	1.0	7.0	0.005640	1.590381
10	B	2	470k	0.164	1.0	4.0	0.003223	1.630664
10	B	3	1M	0.297	1.0	5.0	0.004028	1.600049
10	B	4	470k	0.018	1.0	15.0	0.012085	1.636304
50	A	1	470k	0.121	1.0	2.0	0.001611	1.639929
50	A	2	1M	0.043	1.0	5.0	0.004028	1.620996
50	A	3	470k	0.101	1.0	2.0	0.001611	1.638721
50	A	4	1M	0.000	1.0	7.0	0.005640	1.629053
50	B	1	1M	0.255	1.0	5.0	0.004028	1.644360
50	B	2	470k	0.000	1.0	0.0	0.000000	1.650000
50	B	3	1M	0.125	1.0	1.0	0.000806	1.645569
50	B	4	470k	0.035	1.0	1.0	0.000806	1.650806

Integrity, findings and scope

All 30 captures validated. 1,746,140 received full sweeps (87,307,000 channel values), including warm-up. Firmware acquired 1,746,173 sweeps, sent 1,746,140 and intentionally discarded 33 complete sweeps (0.00189%). Every transmitted sweep was received. No malformed/partial frames, invalid ADC codes or new transfer, data or USB-write errors.

Firmware counter-derived acquisition rates over the full run (including warm-up) differ from the measured delivered rates by at most 0.0142%. These rate estimates use acquired sweep count divided by the recorded full-run duration plus one typical sweep period. USB omissions were too few to materially alter this comparison; individual estimates are in capture_metrics.csv.

Every capture was decoded by an independent structured frame reader and the existing incremental protocol parser using chunk boundaries that split frames. Samples, average timing, start/end timestamps and counts agreed. Device times were unwrapped and checked for strict progression. SHA-256 hashes bind every raw capture to its session metadata.

Settle us	VMID-on rate cost	A spread mV: off / on	B spread mV: off / on
1	50.0%	53.17 / 26.59	146.63 / 74.93
2	49.3%	53.98 / 26.59	128.10 / 71.70
5	48.0%	54.79 / 27.39	97.49 / 62.84
10	46.5%	53.17 / 27.39	68.48 / 52.37
50	44.0%	30.62 / 21.75	9.67 / 9.67

Increasing settling from 1 to 50 us with VMID off lowers received rate from 7,518.1 to 1,091.4 samples/s/channel. In array B, the corresponding baseline spread changes from 146.63 to 9.67 mV. These are measurements under the present open-connector setup, not an electrical diagnosis.

Largest median quiet SD: route 2:4:0, 1 us / VMID off: 0.891 counts (0.000718 V), P-P 8.0 counts (0.006445 V). Channel-specific variations are retained by the common burst mask.

Firmware sampling intervals above 1 ms: 29,316. Long sweep periods are expected at high settling delay; this counter alone is not an error indication. Maximum sampling periods and individual repetition metrics are retained in capture_metrics.csv.

No firmware-source, wire-format or hardware changes were made. The board was left stopped and its original full-array, MUX-order, blocking, 16 MHz, repeat-1, 5 us, VMID-off configuration restored. These captures do not establish ghosting, crosstalk under stimulation, channel identity by known-voltage stimulation, or a minimum adequate settling time for attached sensors.

Evidence: session.json and commands.log; 30 raw captures with status snapshots and hashes; 30 quiet-bin CSVs; capture_metrics.csv; 1,500 per-channel repetition rows; 500 channel summaries; 100 array/MUX group summaries. Reproduce with scripts/capture_adc124_settling.py and scripts/report_adc124_settling.py.